



THE DECISION FRAMEWORK

When not to use Postgres

Default to Postgres, until you hit one of these four walls. Each one has a number.

5 datastores -> 1 engine

A field-tested rule for architects and the leaders who choose the stack: default to Postgres, and make every exit justify itself with a number.

Swipe ->

5 -> 1

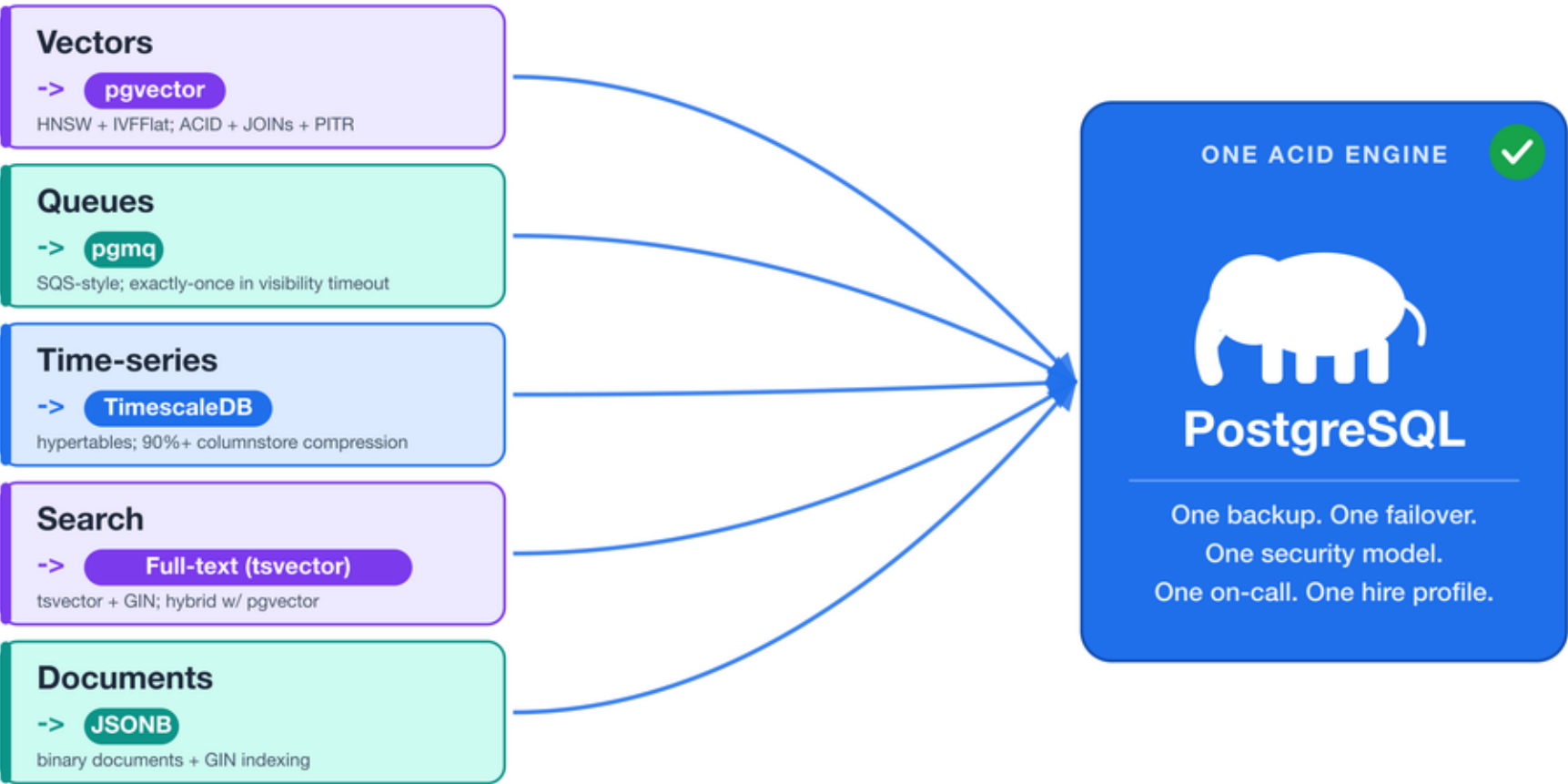
One team ran five datastores to serve one product. pgvector, pgmq, TimescaleDB, full-text, JSONB: it all collapses into one Postgres engine.

Five systems, one engine

Five specialist datastores collapse into one ACID Postgres.

5 -> 1

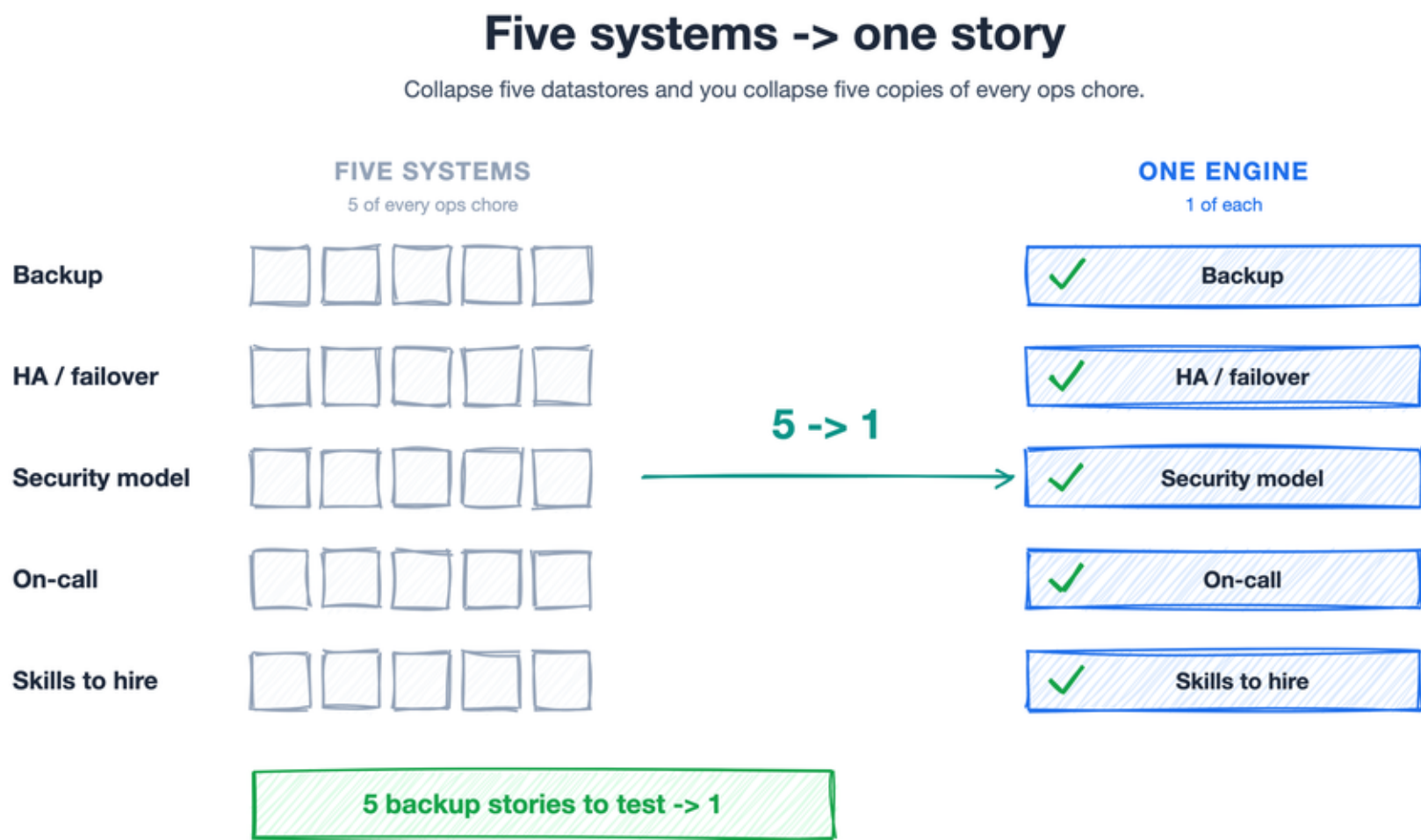
ONE ENGINE



PostgreSQL 18 async I/O -> up to 3x read throughput. Source: reference-brief.md A-B; verified 2026-06-26.

THE OPS DIVIDEND

Five backup stories and five things to patch and get paged about become one backup, one failover, one on-call. The cloud bill drops with them.



Every datastore you don't add is an incident class you never have.

Source: just-use-postgres.md (consolidation dividend); qualitative by design.

THE FOUR WALLS

Postgres isn't always the answer. Four breakpoints justify a specialist, each named with a number, not a vibe.

Where the belief stops -- the four walls

Four workloads where I leave Postgres on purpose -- each justified by one ceiling number.

1. Extreme write throughput

7.5M inserts/sec

@ 4 ms P99 (ScyllaDB, vendor; 2-5x Apache Cassandra)

Cassandra / ScyllaDB

the legitimate specialist exit

Why not PG: a single primary funnels all writes through one node.

Azure: Managed Cassandra / Cosmos DB

2. Planet-scale sharding

ALL of YouTube, 5+ yrs

Vitess ran YouTube's entire DB traffic (also Slack, Square, JD.com)

Spanner / CockroachDB / Vitess

the legitimate specialist exit

Why not PG: no transparent native sharding -- you bolt it on.

Azure: Cosmos DB / PostgreSQL Elastic Clusters

3. Sub-millisecond key-value

single-digit ms

500k+ req/s, 200 TB+, 99.999% (DynamoDB, vendor)

Redis / DynamoDB

the legitimate specialist exit

Why not PG: connection/transaction overhead can't hold that floor.

Azure: Managed Redis / Cosmos DB

4. True OLAP at petabyte scale

~1 billion rows/sec

100M rows in 92 ms, ~7 GB/s per query (ClickHouse)

ClickHouse / Snowflake

the legitimate specialist exit

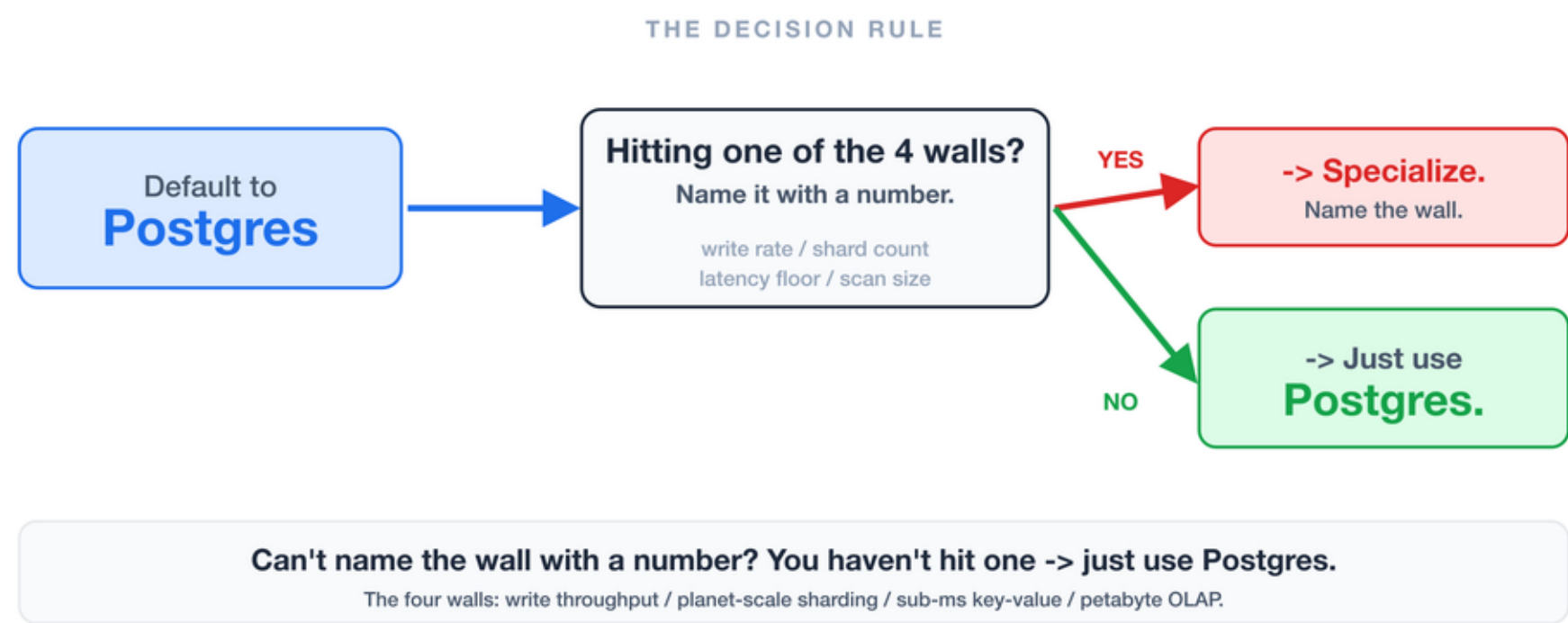
Why not PG: row-oriented OLTP can't match a column store's scan rate.

Azure: Data Explorer (Kusto) / Fabric

Specialists are legitimate equals; Azure is one cloud's option among equals (AWS / GCP / self-hosted equally valid).
Vendor / self-reported figures. Source: reference-brief.md C; verified 2026-06-26.

THE DECISION RULE

Default to Postgres. Hit a wall? Name it with a number. If you can't name the number, you don't have the wall yet.



Source: just-use-postgres.md (the decision rule); verified 2026-06-26.

Run the audit this week

For every datastore beyond Postgres, can your team name the wall it clears and the number? The ones that can't are consolidation candidates.

READ THE FULL ARC

<https://sendtoshailsh.github.io/content-creation/blog/just-use-postgres.html>